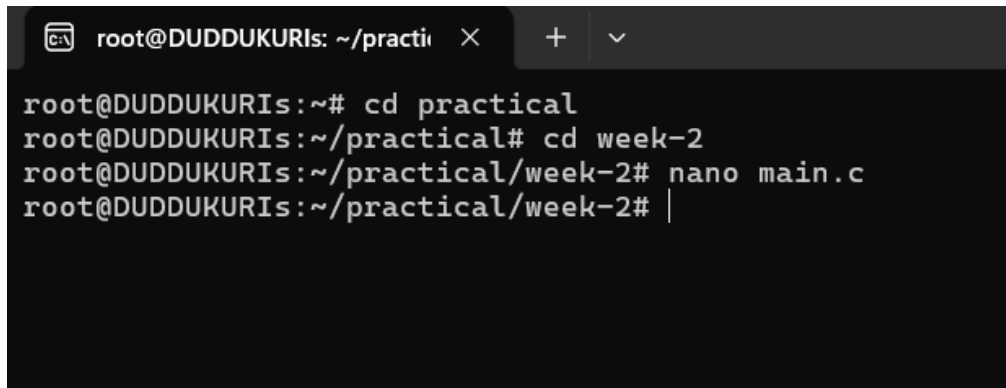

PRACTICAL WEEK 2

Task 1: Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

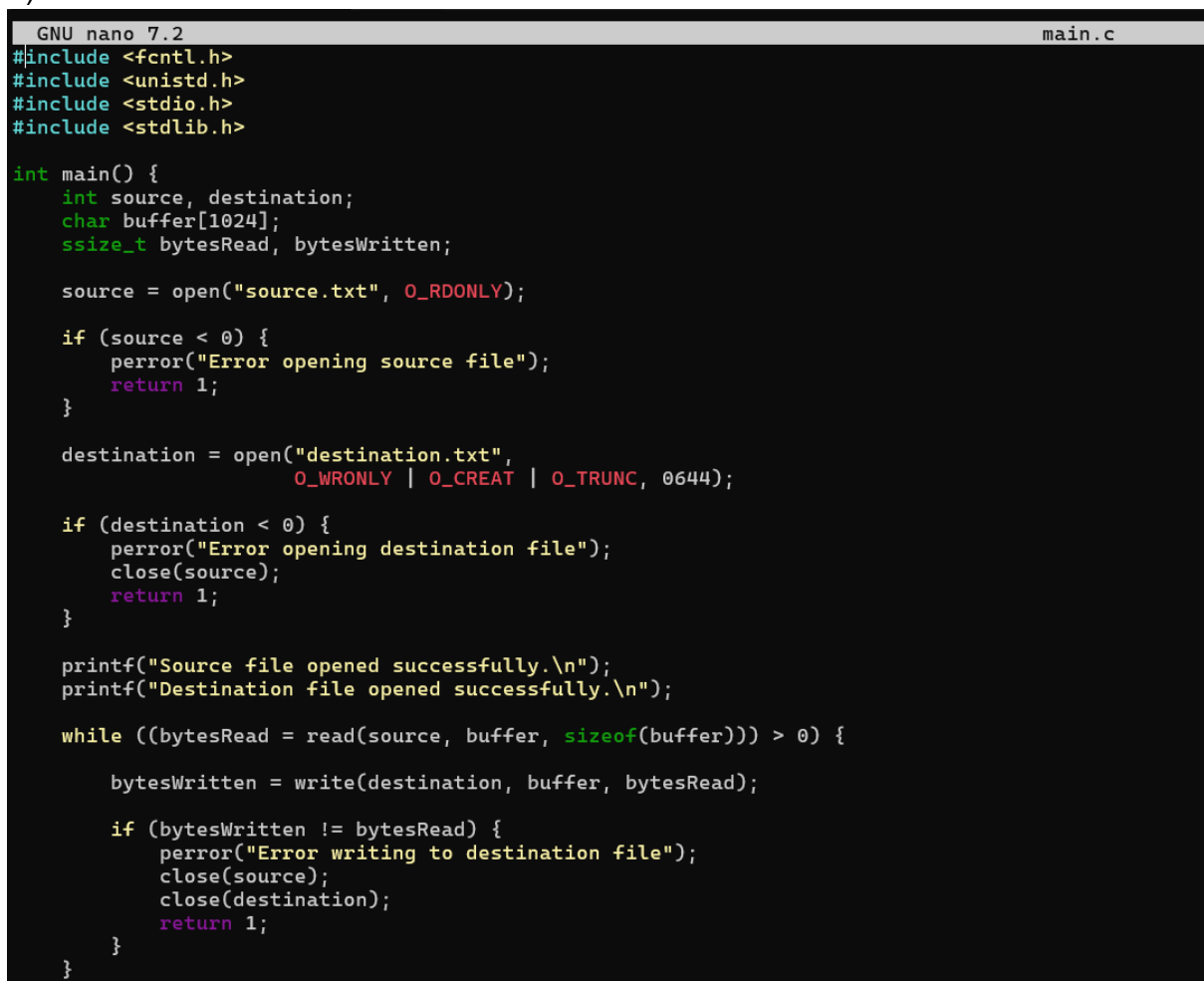
1) Create a file

A terminal window with a dark background. The title bar shows 'root@DUDDUKURIs: ~/practi' and a tab icon. The terminal text shows the user navigating to the 'practical' directory, then 'week-2', and using 'nano' to create a file named 'main.c'.

```
root@DUDDUKURIs:~# cd practical
root@DUDDUKURIs:~/practical# cd week-2
root@DUDDUKURIs:~/practical/week-2# nano main.c
root@DUDDUKURIs:~/practical/week-2# |
```

Then press enter to open command

2) write a code

A terminal window showing the GNU nano 7.2 editor. The title bar shows 'GNU nano 7.2' and 'main.c'. The code is a C program that opens 'source.txt' in read-only mode and 'destination.txt' in write-only mode with O_CREAT and O_TRUNC flags. It then reads from the source file into a buffer and writes it to the destination file until the source file is exhausted. Error handling is included for both opening and writing.

```
GNU nano 7.2 main.c
#include <fcntl.h>
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>

int main() {
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead, bytesWritten;

    source = open("source.txt", O_RDONLY);

    if (source < 0) {
        perror("Error opening source file");
        return 1;
    }

    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0) {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    printf("Source file opened successfully.\n");
    printf("Destination file opened successfully.\n");

    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0) {
        bytesWritten = write(destination, buffer, bytesRead);

        if (bytesWritten != bytesRead) {
            perror("Error writing to destination file");
            close(source);
            close(destination);
            return 1;
        }
    }
}
```

```

        close(destination);
        return 1;
    }
}

if (bytesRead < 0) {
    perror("Error reading source file");
} else {
    printf("File contents copied successfully.\n");
}

close(source);
close(destination);

printf("Files closed successfully.\n");

return 0;
}

```

- 1) To save press control O
- 2) To come back to shell press control X
- 3) Then DO compilation

```

root@DUDDUKURIs:~# cd practical
root@DUDDUKURIs:~/practical# cd week-2
root@DUDDUKURIs:~/practical/week-2# nano main.c
root@DUDDUKURIs:~/practical/week-2# gcc main.c -o main
root@DUDDUKURIs:~/practical/week-2# |

```

OUTPUT:

```

root@DUDDUKURIs: ~/practi  ×  +  v
root@DUDDUKURIs:~# cd practical
root@DUDDUKURIs:~/practical# cd week-2
root@DUDDUKURIs:~/practical/week-2# nano main.c
root@DUDDUKURIs:~/practical/week-2# gcc main.c -o main
root@DUDDUKURIs:~/practical/week-2# echo "This is a sample file for system call demonstration." > source.txt
root@DUDDUKURIs:~/practical/week-2# cat source.txt
This is a sample file for system call demonstration.
root@DUDDUKURIs:~/practical/week-2# ./main.c
-bash: ./main.c: Permission denied
root@DUDDUKURIs:~/practical/week-2# ./main
Source file opened successfully.
Destination file opened successfully.
File contents copied successfully.
Files closed successfully.
root@DUDDUKURIs:~/practical/week-2# |

```

Task-2: Use the strace utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

```
root@DUDDUKURIS:~/practical/week-2#
root@DUDDUKURIS:~/practical/week-2# echo "Hello, this is Task 2 system call demonstration." > sample2.txt
root@DUDDUKURIS:~/practical/week-2# cat sample2.txt
Hello, this is Task 2 system call demonstration.
root@DUDDUKURIS:~/practical/week-2# |
```

```
root@DUDDUKURIS:~/practical/week-2#
root@DUDDUKURIS:~/practical/week-2# strace -e trace=openat,read,write,close,execve,exit_group cat sample2.txt
execve("/usr/bin/cat", ["cat", "sample2.txt"], 0x7ffdf4f429c8 /* 26 vars */) = 0
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0\220\243\2\0\0\0\0"... , 832) = 832
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/locale-archive", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/share/locale/locale.alias", O_RDONLY|O_CLOEXEC) = 3
read(3, "# Locale name alias data base.\n#\n...", 4096) = 2996
read(3, "", 4096) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_IDENTIFICATION", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MEASUREMENT", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TELEPHONE", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_ADDRESS", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NAME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NAME", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_PAPER", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_PAPER", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_MONETARY", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_COLLATE", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TIME", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_TIME", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_TIME", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "sample2.txt", O_RDONLY) = 3
read(3, "Hello, this is Task 2 system cal"... , 131072) = 49
write(1, "Hello, this is Task 2 system cal"... , 49)Hello, this is Task 2 system call demonstration.
) = 49
read(3, "", 131072) = 0
close(3) = 0
close(1) = 0
close(2) = 0
exit_group(0) = ?
+++ exited with 0 +++
root@DUDDUKURIS:~/practical/week-2# |
```